Econ 470 HW 4.3

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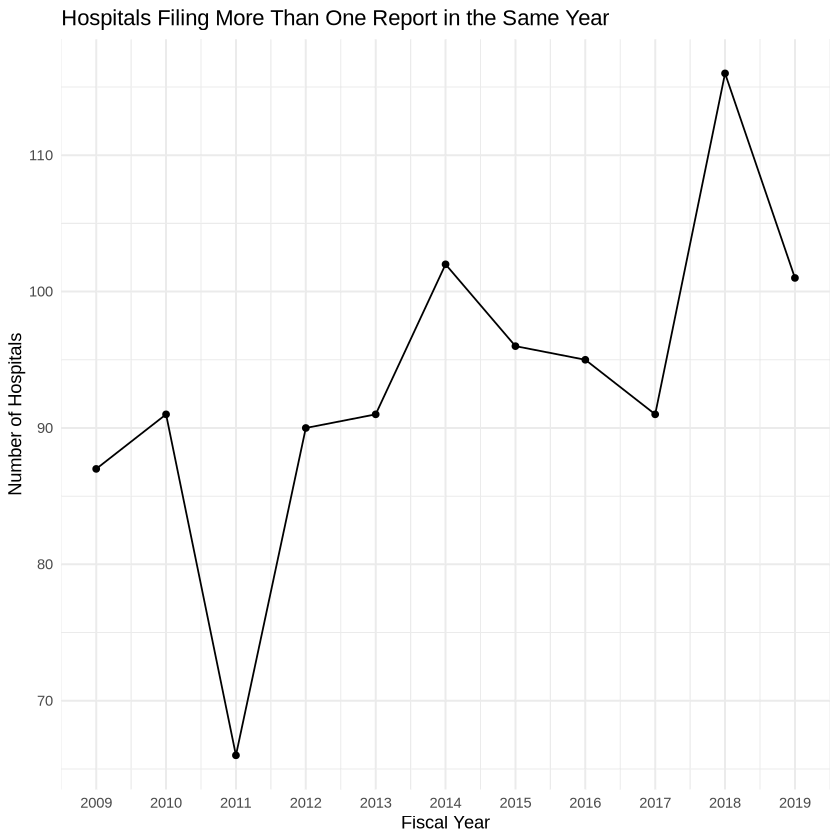
Question 1



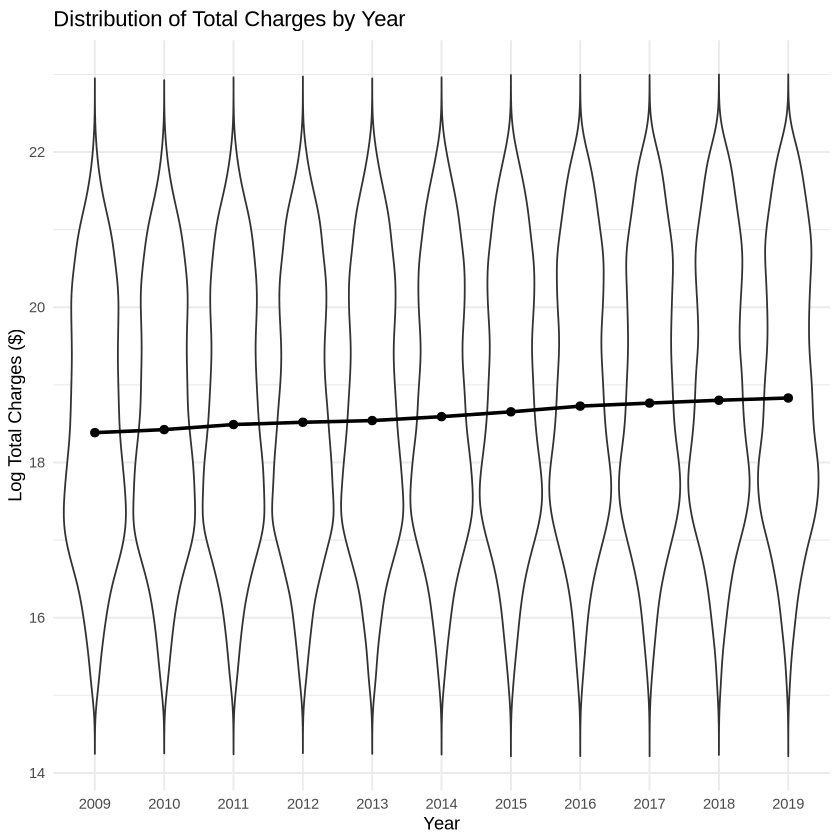
Question 2

A tibble: 11 × 2

Year	Unique Hospital IDs
<dbl>	<int>
2009	6100
2010	6103
2011	6097
2012	6140
2013	6066
2014	6064
2015	6051
2016	6091
2017	6083
2018	6042
2019	6031

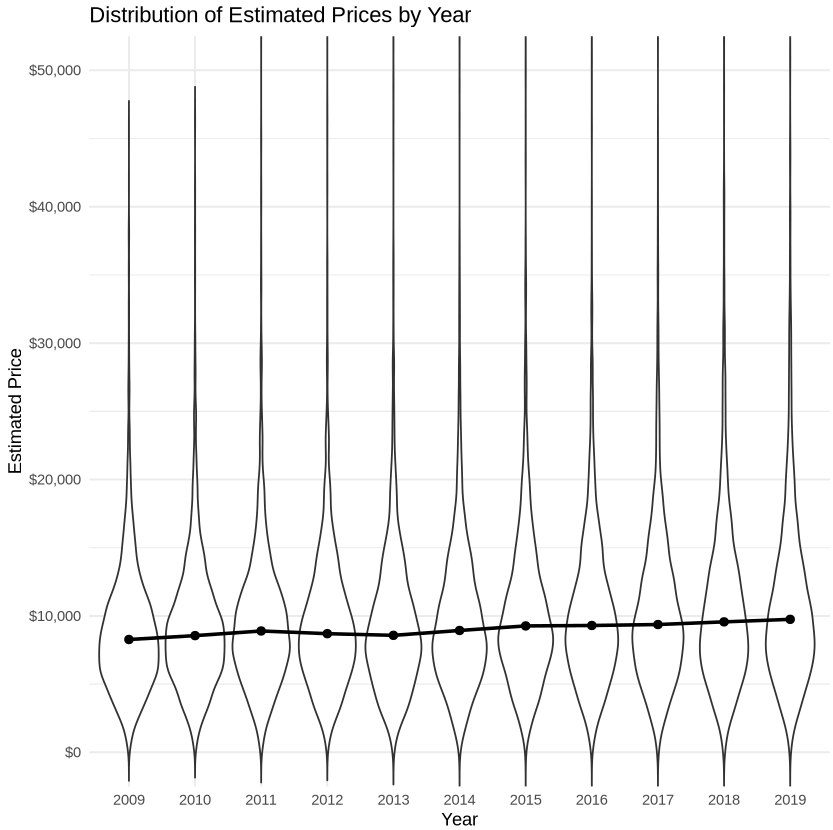
After removing/combining multiple reports, slightly more than 6,000 unique hospital IDs exist in the data each year.

Question 3



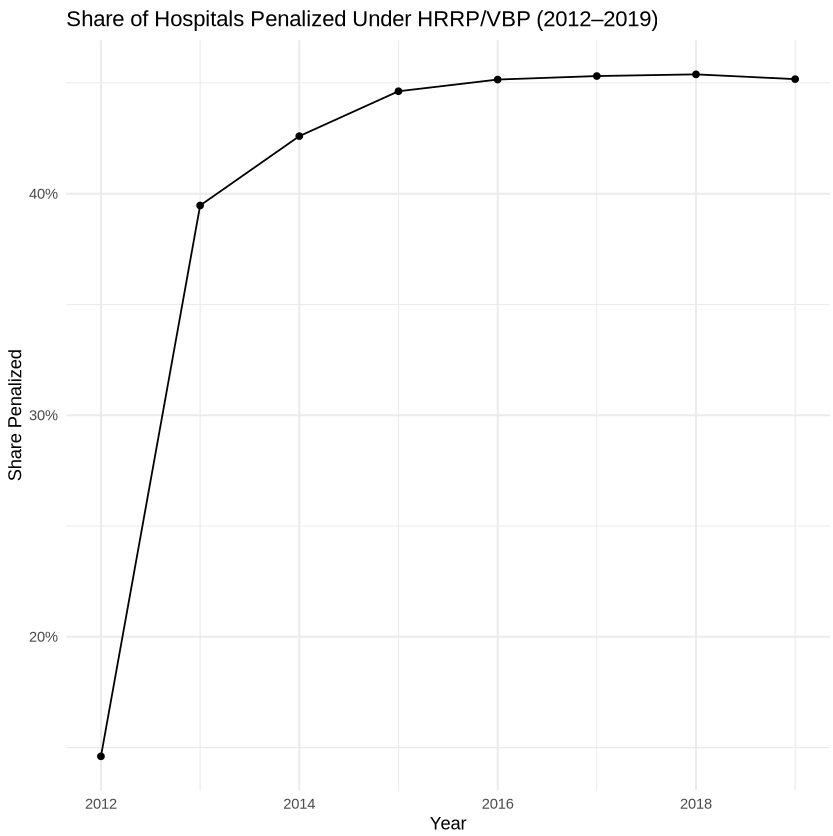
Total charges increase gradually over time from 2009 to 2019.

Question 4



Estimated prices increase gradually over time, and the median price is around $10,000 in 2019.

Question 5



Estimate ATEs

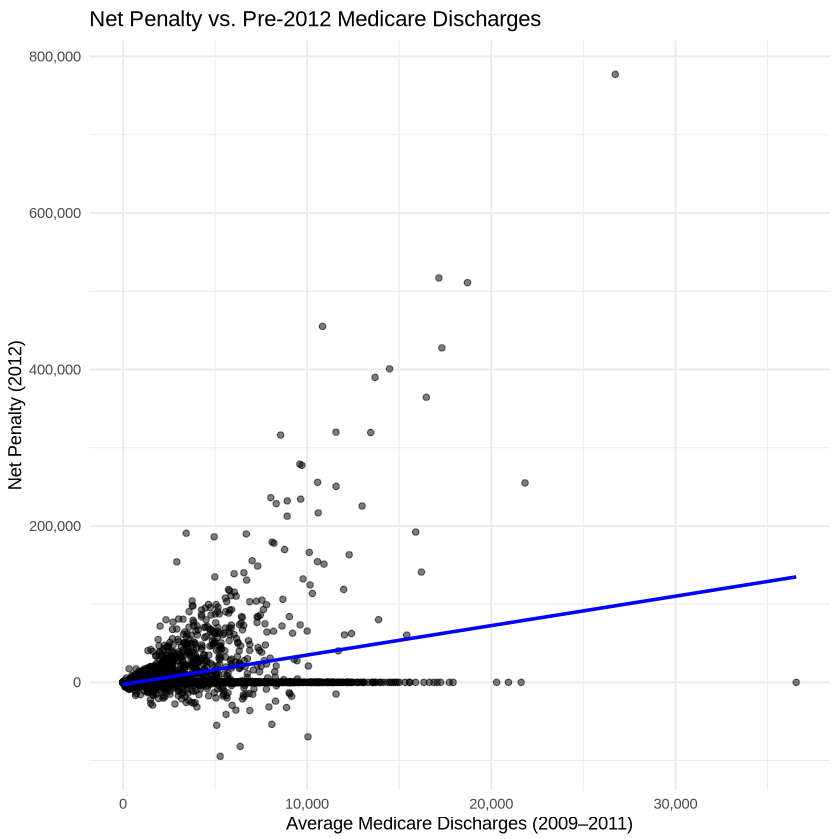
Question 6

OLS Estimates of the Effect of 2012 Net Penalties on Price Changes (2011–2014)

	Dependent variable:		
	Price Change		
	BL	BL + Beds	BL + Beds + Medicaid
	(1)	(2)	(3)
Net Penalty (2012)	0.003	0.003	0.002
	(0.009)	(0.009)	(0.010)
Mean Beds (2009–2011)		0.117	0.103
		(0.239)	(0.240)
Mean Medicaid Discharges (2009–2011)			0.135
			(0.175)
Constant	724.252*	696.155*	487.229
	(380.804)	(385.199)	(471.688)
Observations	2,454	2,454	2,454
R2	0.00004	0.0001	0.0004
Adjusted R2	-0.0004	-0.001	-0.001
Note:			*p<0.1; **p<0.05; ***p<0.01

There is no statistically significant effect of net penalties on price changes for any of the OLS models.

Question 7



Higher Medicare discharges (2009-2011) are associated with higher 2012 net penalties, on average.

Question 8

First Stage Estimates

	Dependent variable:		
	Net Penalty (2012)		
	BL	BL + Beds	BL + Beds + Medicaid
	(1)	(2)	(3)
Avg Medicare Discharges (2009–2011)	4.181***	4.182***	4.591***
	(0.237)	(0.238)	(0.273)
Mean Beds (2009–2011)		-0.027	0.013
		(0.483)	(0.483)
Mean Medicaid Discharges (2009–2011)			-1.227***
			(0.401)
Constant	-5,839.040***	-5,837.246***	-5,329.239***
	(1,138.931)	(1,139.608)	(1,149.739)
Observations	2,454	2,454	2,454
R2	0.113	0.113	0.116
Adjusted R2	0.112	0.112	0.115

Note: *p<0.1; **p<0.05; ***p<0.01

Reduced Form Estimates

	Dependent variable:		
	Price Change (2011–2014)		
	BL	BL + Beds	BL + Beds + Medicaid
	(1)	(2)	(3)
Avg Medicare Discharges (2009–2011)	0.275**	0.271**	0.296**
	(0.118)	(0.118)	(0.136)
Mean Beds (2009–2011)		0.061	0.064
		(0.240)	(0.240)
Mean Medicaid Discharges (2009–2011)			-0.075
			(0.200)
Constant	-247.081	-251.124	-220.085
	(565.771)	(566.100)	(572.205)
Observations	2,454	2,454	2,454
R2	0.002	0.002	0.002
Adjusted R2	0.002	0.001	0.001

Note: *p<0.1; **p<0.05; ***p<0.01

The first-stage results show a statistically significant positive relationship between pre-2012 Medicare discharges and 2012 net penalties, meaning higher Medicare discharges are associated with higher penalties. The reduced-form estimates show that higher Medicare discharges is associated with increased price growth from 2011 to 2014.

Question 9

IV Estimates of the Effect of Net Penalties on Price Changes (2011–2014)

	Dependent variable:		
	Price Change		
	BL	BL + Beds	BL + Beds + Medicaid
	(1)	(2)	(3)
Net Penalty (2012)	0.066**	0.065**	0.065**
	(0.028)	(0.029)	(0.030)
Mean Beds (2009–2011)		0.063	0.063
		(0.242)	(0.243)
Mean Medicaid Discharges (2009–2011)			0.004
			(0.186)
Constant	136.455	127.745	124.003
	(458.821)	(460.088)	(503.298)
Observations	2,454	2,454	2,454
R2	-0.018	-0.017	-0.017
Adjusted R2	-0.018	-0.018	-0.019
Note:			*p<0.1; **p<0.05; ***p<0.01

The IV estimates indicate that higher net penalties are associated with increased prices, and the results are statistically significant. The IV estimator helps to correct for bias due to the endogeneity of penalty status.

Question 10

The IV estimate provides a Local ATE, which shows the effect of penalties on price changes for hospitals that have penalties that are influenced by their pre-2012 Medicare discharges. The LATE may differ from the overall ATE because it only applies to hospitals whose penalty status is affected by the instrument (prior Medicare discharges), rather than all hospitals. Thus, it may not generalize to all hospitals (for instance, those with very low Medicare discharges).